MILITARY SPECIFICATION SHEET

RELAYS, ELECTROMAGNETIC, ESTABLISHED RELIABILITY, DPDT,
LOW LEVEL TO 10 AMPERES, HERMETICALLY SEALED, ALL WELDED, DC COILS
(0.200 INCH TERMINAL SPACING, .050 DIAMETER TERMINAL PINS)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-R-83536.

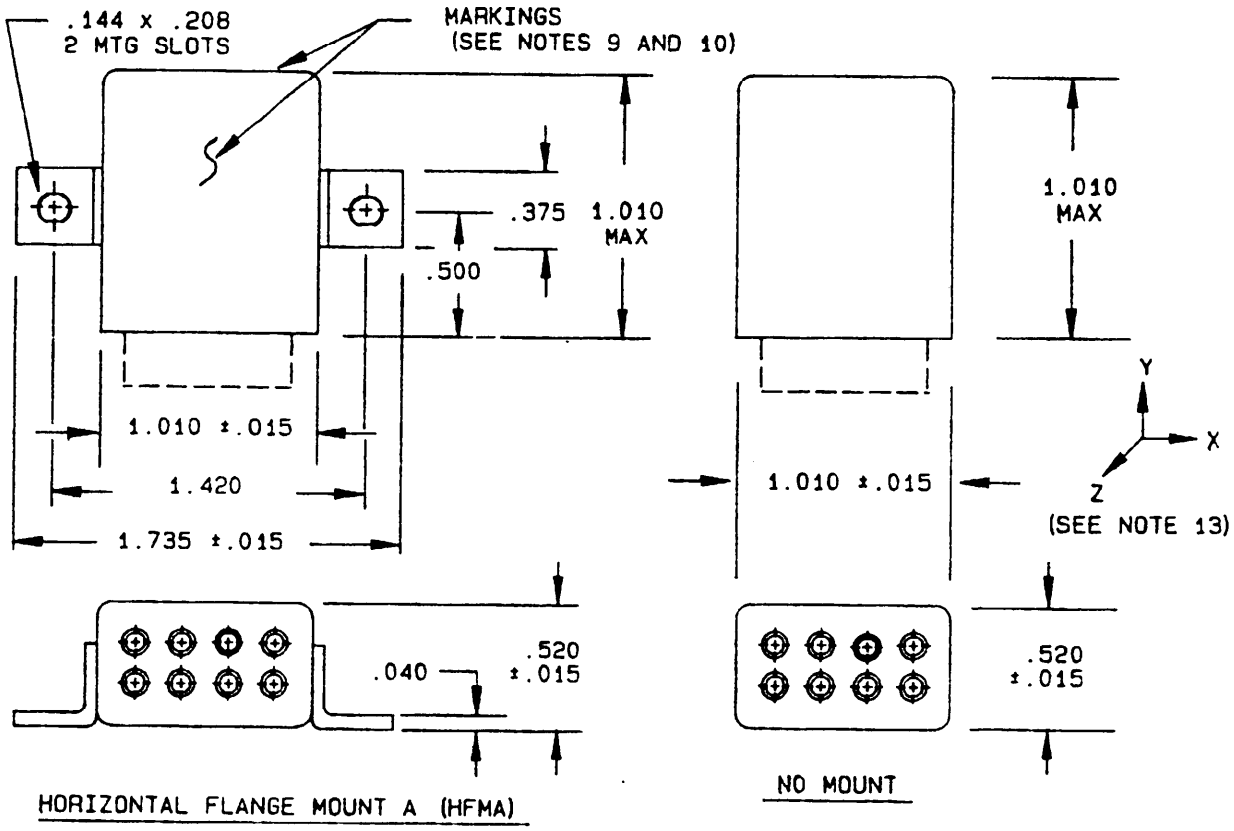
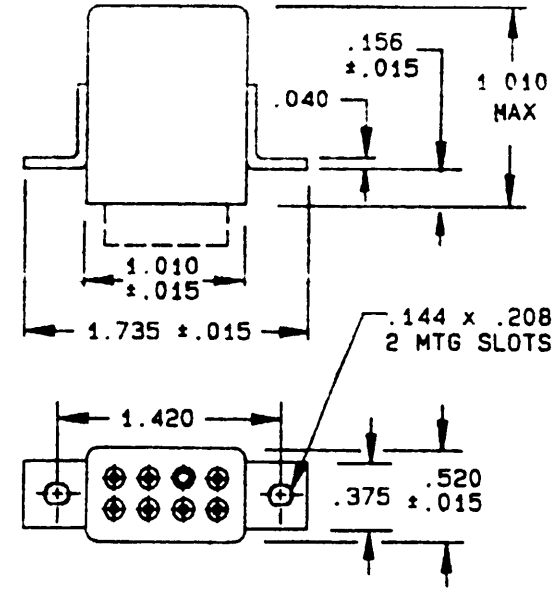


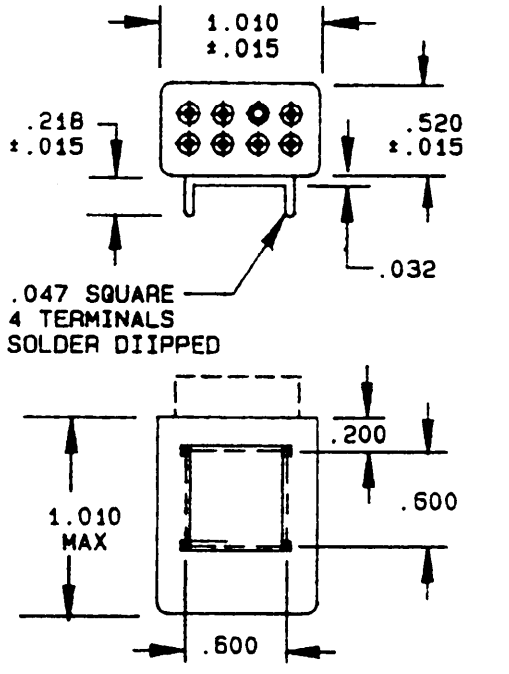
FIGURE 1. Dimensions and configurations.

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RAISED VERTICAL FLANGE MOUNT (RVFM)

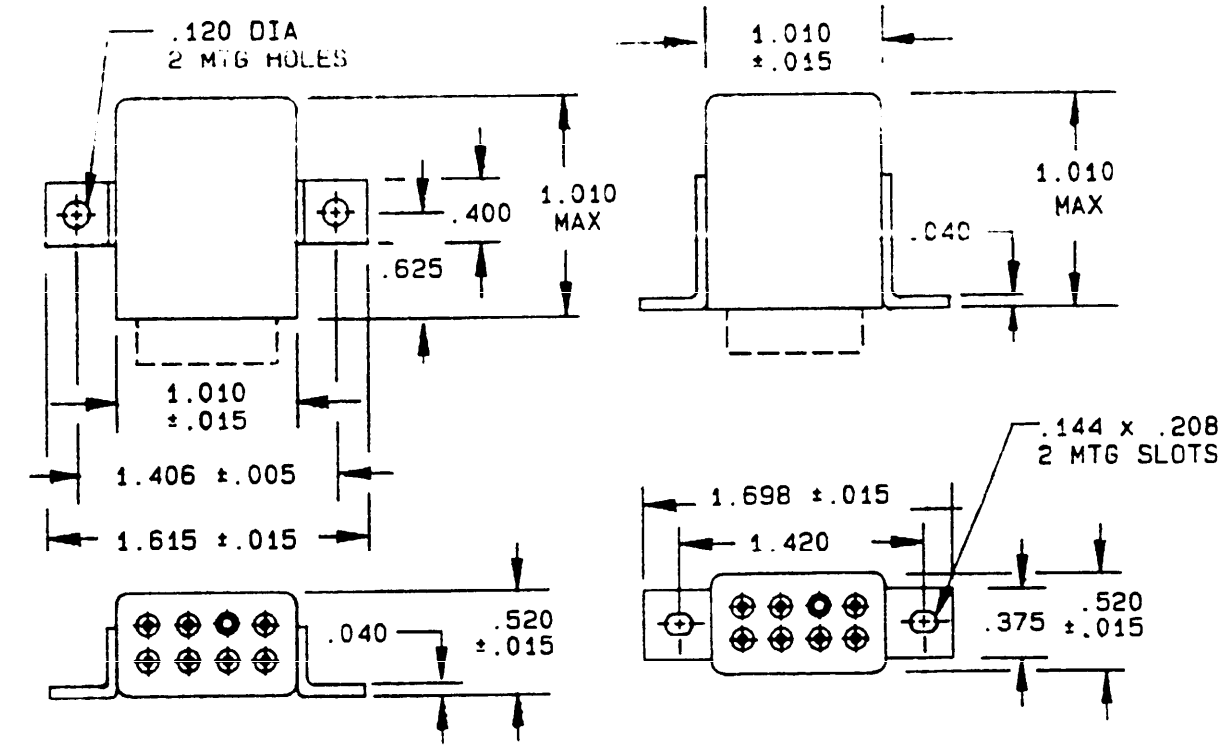


LOW PROFILE MOUNT (LPM)

(LP TERMINAL CONFIGURATION ONLY)

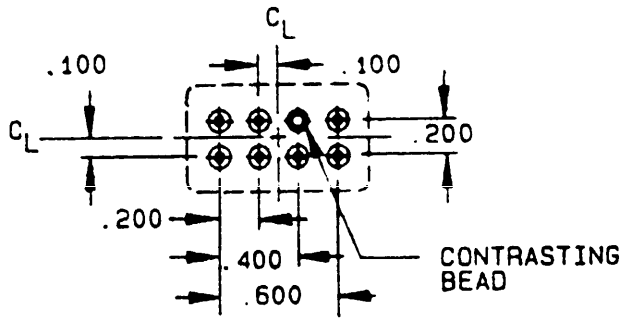
FIGURE 1. Dimensions and configurations - Continued.

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HORIZONTAL FLANGE MOUNT B (HFMB)

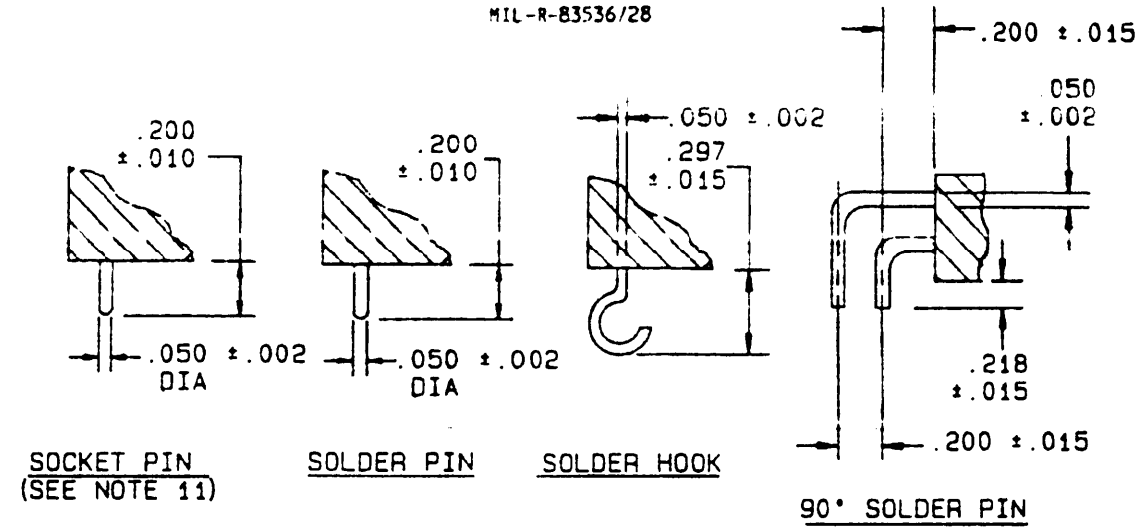
VERTICAL FLANGE MOUNT (VFM)



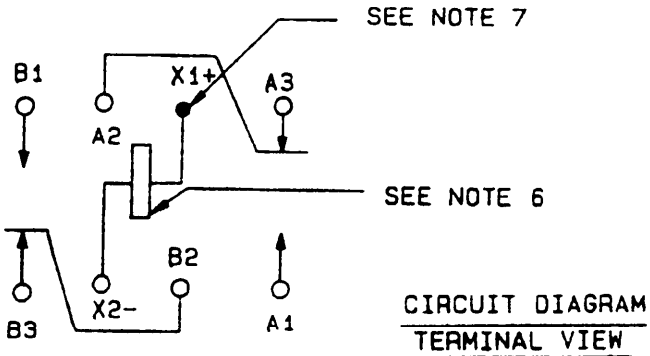
TERMINAL DIMENSIONS (TYPICAL)

FIGURE 1. Dimensions and configurations - Continued.

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TERMINALS



CIRCUIT DIAGRAM
TERMINAL VIEW
DEENERGIZED POSITION

Inches	mm	Inches	mm
.002	0.05	.297	7.54
.005	0.13	.375	9.52
.010	0.25	.400	10.16
.015	0.38	.500	12.70
.032	0.81	.520	13.21
.040	1.02	.600	15.24
.047	1.19	.625	15.88
.050	1.27	1.010	25.65
.100	2.54	1.250	31.75
.120	3.05	1.406	35.71
.144	3.66	1.420	36.07
.156	3.96	1.615	41.02
.200	5.08	1.698	43.13
.208	5.28	1.735	44.07
.218	5.54		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.010 (0.25 mm).
4. Terminal locating dimensions shown are applicable to all type mounts.
5. The shape and orientation of hook terminals is optional. Hook terminals to accommodate three number 22 AWG wires.
6. Coil symbol is optional in accordance with MIL-STD-1285.
7. Indicated terminal shall be identified with a contrasting bead.
8. There shall be affixed to the relay a suitable legible circuit diagram that identifies each terminal location specified.
9. Applicable to horizontal flange mount only. The circuit diagram, manufacturer's PIN, and the military PIN shall be marked on the near side. The remaining portion of the nameplate data shall be marked on the far side.
10. Applicable to horizontal flange mount only. Relays shall be marked with the manufacturer's name or CAGE code and date code. Marking shall be with the bottom of the print adjacent to the near side.
11. Socket pin terminals, except the polarizing pin, shall be gold-plated in accordance with MIL-G-45204, type II, class I. The nickel underplating shall be 50 to 150 microinches thick minimum.
12. 90° solder pin terminals are for use with low profile mount and horizontal flange mount A enclosure type only.
13. Plane of critical motion for vibration and shock is Z-axis.

FIGURE 1. Dimensions and configurations - Continued.

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REQUIREMENTS:

Contact data:

Load ratings: See table I.

Low level: 10 to 50 μA at 10 to 50 mV dc or peak ac.

Intermediate current: Applicable.

Mixed loads: Applicable.

TABLE I. Rated contact load and cycles (amperes per pole).

Type of load (high level)	Cycles x 10^3	28 V dc	115 V ac 1 phase 400 Hz	115/200 V ac 3 phase 400 Hz	115 V ac 60 Hz 1/
Resistive	100	10	10	10	3
Inductive	100	4	4	4	2
Motor 2/	100	3	3	3	1.5
Lamp	100	2	2	2	0.5

1/ Life cycle rating 50,000 cycles.
2/ The relay shall be subjected to the specified life cycles for making five times the rated motor load at rated voltage, and breaking the normal rated load. Duration of the inrush current shall be 0.07 ±0.02 second, after which it shall be reduced to its rated motor load for the remainder of the on-time. Cycling rate shall be 0.35 ±0.09 second on and 2.0 ±0.1 seconds off.

Life: 100,000 cycles, unless otherwise specified (see table I).

Contact voltage drop or resistance:

Initial contact voltage drop: 0.100 volt maximum. Rated resistive current at 6 V dc or peak ac. Relays shall not make or break this load.

Initial contact resistance: 0.010 ohm maximum. 50 mA maximum at 6 V dc or peak ac.

High level life (contact voltage drop):

During life: 10 percent open circuit voltage maximum.

After life: 0.200 volt maximum.

Intermediate current (contact resistance):

During intermediate current: 3 ohms maximum.

After intermediate current: 1 ohm maximum.

Low level (contact resistance):

During low level: 100 ohms maximum.

After low level: 0.150 ohm maximum at 100 mA and 28 V dc.

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Contact bounce.

N/O: 2.0 ms maximum.

N/C: 5.0 ms maximum.

Contact stabilization time:

N/O: 2.5 ms maximum.

N/C: 5.5 ms maximum.

Overload current:

Resistive: 25 amperes at 28 V dc and 115 V ac, 400 Hz.

Inductive: Twice rated current, ac or dc.

Coil data 1/: See table II.

Operate time: 15 milliseconds maximum with rated coil voltage at 25°C. 15 milliseconds maximum with rated coil voltage over the temperature range.

Release time: 15 milliseconds maximum from rated coil voltage at 25°C. 15 milliseconds maximum from rated coil voltage over the temperature range.

Duty rating: Continuous.

Coil transient suppression: Not applicable.

Neutral screen: Not applicable.

Electrical data:

Insulation resistance:

Initial: 100 megohms minimum at 500 V dc.

After life or environmental tests: 50 megohms at 500 V dc.

1/ CAUTION: Due to possible interaction of relay magnetic fields, the following spacing requirements, as a minimum, shall be considered in dense packaging situations:

a. Row to row assisting fields: .125 (3.18 mm).

b. Row to row opposing fields: .1875 (4.763 mm).

c. Side to side alternating fields: .0625 (1.588 mm).

d. Side to side like fields: .125 (3.18 mm).

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TABLE II. Dash numbers and characteristics. 1/

Dash number 2/ M83536/28-				Mount	Coil voltage (V dc) 3/		At 25°C				Over temperature range		
Socket pin	Solder pin	Solder hook	90° solder pin		Rated	Max	Coil resistance (ohms) minimum	Specified pickup value (voltage) (V dc) 4/	Specified hold value (voltage) (V dc) 4/	Specified dropout value (voltage) (V dc) 4/	Specified pickup value (voltage) (V dc) 4/	Specified hold value (voltage) (V dc) 4/	Specified dropout value (voltage) (V dc) 4/
---	001	002	003	HFM A	6.0	9.0	14.0	3.7	1.79	.40	5.0	2.5	.25
004	005	006	---	RVFM									
---	---	---	007	LPM									
---	008	009	---	HFM B									
010	011	012	---	VFM									
013	014	---	---	No mount									
---	015	016	017	HFM A	12.0	16.0	58	7.4	3.6	.80	10.0	5.0	.50
018	019	020	---	RVFM									
---	---	---	021	LPM									
---	022	023	---	HFM B									
024	025	026	---	VFM									
027	028	---	---	No mount									
---	029	030	031	HFM A	18.0	24.0	153	11.0	5.4	1.2	15.0	7.5	.75
032	033	034	---	RVFM									
---	---	---	035	LPM									
---	036	037	---	HFM B									
038	039	040	---	VFM									
041	042	---	---	No mount									

See footnotes at end of table.

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TABLE II. Dash numbers and characteristics - Continued. 1/

Dash number M83536/28- 2/				Mount	Coil voltage (V dc) 3/		At 25°C				Over temperature range		
Socket pin	Solder pin	Solder hook	90° solder pin		Rated	Max	Coil resistance (ohms) minimum	Specified pickup value (voltage) (V dc) 4/	Specified hold value (voltage) (V dc) 4/	Specified dropout value (voltage) (V dc) 4/	Specified pickup value (voltage) (V dc) 4/	Specified hold value (voltage) (V dc) 4/	Specified dropout value (voltage) (V dc) 4/
---	043	044	045	HFM A	28.0	32.0	270	14.7	7.1	1.8	20.0	10.0	1.0
046	047	048	---	RFVM									
---	---	---	049	LPM									
---	050	051	---	HFM B									
052	053	054	---	VFM									
055	056	---	---	No mount									
---	057	058	059	HFM A	48.0	52.0	900	29.5	14.3	3.6	40.0	20.0	2.0
060	061	062	---	RVFM									
---	---	---	063	LPM									
---	064	065	---	HFM B									
066	067	068	---	VFM									
069	070	---	---	No mount									
---	071	072	073	HFM A	120	122	6840	74.0	35.8	8.0	100.0	50.0	5.0
074	075	076	---	RVFM									
---	---	---	077	LPM									
---	078	079	---	HFM B									
080	081	082	---	VFM									
083	084	---	---	No mount									

See footnotes on next page.

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TABLE II. Dash numbers and characteristics - Continued. 1/

1/ Each relay possesses high level and low level capabilities. However, relays previously tested or used above 10 mA resistive at 6 V dc maximum or peak ac open circuits are not recommended for subsequent use in low level applications.
2/ The suffix letter L, M, P, or R to designate the applicable failure rate level shall be added to the applicable listed dash number. Failure rate level (percent per 10,000 cycles): L, 3.0; M, 1.0; P, 0.1; R, 0.01. Example: 001L, 002R.
3/ CAUTION: The use of any coil voltage less than the rated coil voltage will compromise the operation of the relay.
4/ Pickup, hold, and dropout voltages as shown are for test purposes only and are not to be used for design criteria.

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Dielectric withstanding voltage: 2/

	Sea level V rms (60 Hz)	Altitude V rms (60 Hz)
Between case, frame, or enclosure, and all contacts in the energized and deenergized position	1,000	500 All terminals to case
Between case, frame, or enclosure and coil	500	
Between all contacts and coil	1,000	
Between open contacts in the energized and deenergized positions	1,000	
Between contact poles	1,000	

Environmental characteristics:

Temperature range: -65°C to +125°C.

Shock (specified pulse): Applicable, MIL-STD-202, method 213, test condition C, except peak value shall be 100 g's for 6 ±1 ms. Contact chatter shall not exceed 10 microseconds maximum for closed contacts and 1 microsecond maximum closure for open contacts.

Vibration (sinusoidal): Applicable, MIL-STD-202, method 204, 20 g's except frequency range shall be 10 to 2,000 Hz. Contact chatter shall not exceed 10 microseconds maximum for closed contacts and 1 microsecond maximum closure for open contacts.

Magnetic interference: Applicable.

Resistance to solvents: Applicable. Temperature of solvents A, B, C, and D shall be maintained between 63°C to 70°C.

Acceleration: 17 g's.

Physical data:

Dimensions and configurations: See figure 1.

Weight: .125 pound (56.7 g's) maximum.

Terminal strength:

Solder hook terminals:

Pull force: 3 pounds ±0.3 pound.

Bend: Applicable.

Twist: Not applicable.

Solder pin terminals:

Pull force: 3 pounds ±0.3 pound.

Bend: Applicable.

Twist: Not applicable.

2/ Dielectric may be improved by suitable insulation of terminals and wiring after installation.

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Socket pin terminals:

Pull force: 3 pounds ±0.3 pound.

Bend: Applicable.

Twist: Not applicable.

Terminal solderability: Applicable to solder pin and solder hook terminals only.

Seal: Hermetic.

Marking: Applicable.

Part or Identifying Number (PIN): M83536/28 (dash number from table II and suffix letter designating failure rate level). PIN is a new term encompassing previous terms used in specifications such as part number, type designator, and identification number.

Qualification inspection:

Qualification inspection and sample size: See table III.

TABLE III. Qualification inspection and sample size.

Single submission	Group submission	
38 units plus 1 open unit for level L at C = 0 1/	M83536/28-046	38 units plus 1 open unit for level L at C = 0 1/ Qualification inspection as applicable.
	M83536/28-043 M83536/28-045 M83536/28-051 M83536/28-049	2 units each part number, qualification inspection table, group II, shock, vibration (sinusoidal), acceleration, terminal strength, and seal.
	M83536/28-001 M83536/28-018 M83536/28-035 M83536/28-053 M83536/28-069 M83536/28-084	2 units, qualification inspection table, group II

1/ The number of units required for qualification testing will be increased as required in group V, table II of MIL-R-83536, if the relay manufacturer elects to test the number of units permitting one or more failures. Prior to performance of qualification testing, the relay manufacturer shall preselect the sampling plan.

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Intermediate current: Applicable.

Qualification inspection (reduced testing) (sample size - 2 units each coil voltage and 1 unsealed unit). See table IV.

If the relays produced for MIL-R-83536/28 are similar in construction and design except for the suppression network to the relays produced for MIL-R-83536/29 then reduced testing for qualification of MIL-R-83536/28 relays may be performed concurrent with or subsequent to successful qualification of MIL-R-83536/29 relays. For reduced testing see table IV.

TABLE IV. Qualification inspection (reduced testing).

Inspection
2 units each coil voltage Group II of qualification inspection table 1 unsealed sample unit for internal inspection

Supersession data:

Supersession data: See table V.

TABLE V. Supersession data.

Superseded PIN M5757/23-	Replacement PIN M83536/28- 1/	Superseded PIN M5757/23-	Replacement PIN M83536/28- 1/
001	048	019	066
002	054	020	069
003	046	026	076
004	052	027	082
005	055	028	074
006	020	029	080
007	026	030	083
008	018	031	051
009	024	032	023
010	027	033	009
011	006	034	065
012	012	036	079
013	004	037	034
014	010	038	040
015	013	039	032

See footnote at end of table.

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TABLE V. Supersession data - Continued.

Superseded PIN M5757/23-	Replacement PIN M83536/28- 1/	Superseded PIN M5757/27-	Replacement PIN M83536/28- 1/
016	062	040	038
017	068	041	041
018	060	042	037
Superseded PIN MS27245-	**Replacement PIN M83536/28- 1/**	**Superseded PIN MS27247-**	**Replacement PIN M83536/28- 1/**
001	048	004	053
003	048	005	053

1/ Complete PIN shall contain suffix letter L, M, P, or R to designate failure rate level (see 2/ of table II). A PIN with any failure rate supersedes the applicable MIL-R-5757, MS27245, and MS27247 part.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Air Force - 99
DLA - ES

Preparing activity:
Air Force - 85

Agent:
DLA - ES

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